

CANDIDATE
NAME

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PDG
INDEX NUMBER

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BIOLOGY

9648/02

Paper 2 Core Paper

18 September 2015
Friday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
Section A	80
1	
2	
3	
4	
5	
6	
7	
8	
Section B	20
9 / 10	
Total	100

Section A

Answer **all** the questions in this section.

- 1 The synthesis of collagen is shown in Fig 1.1.

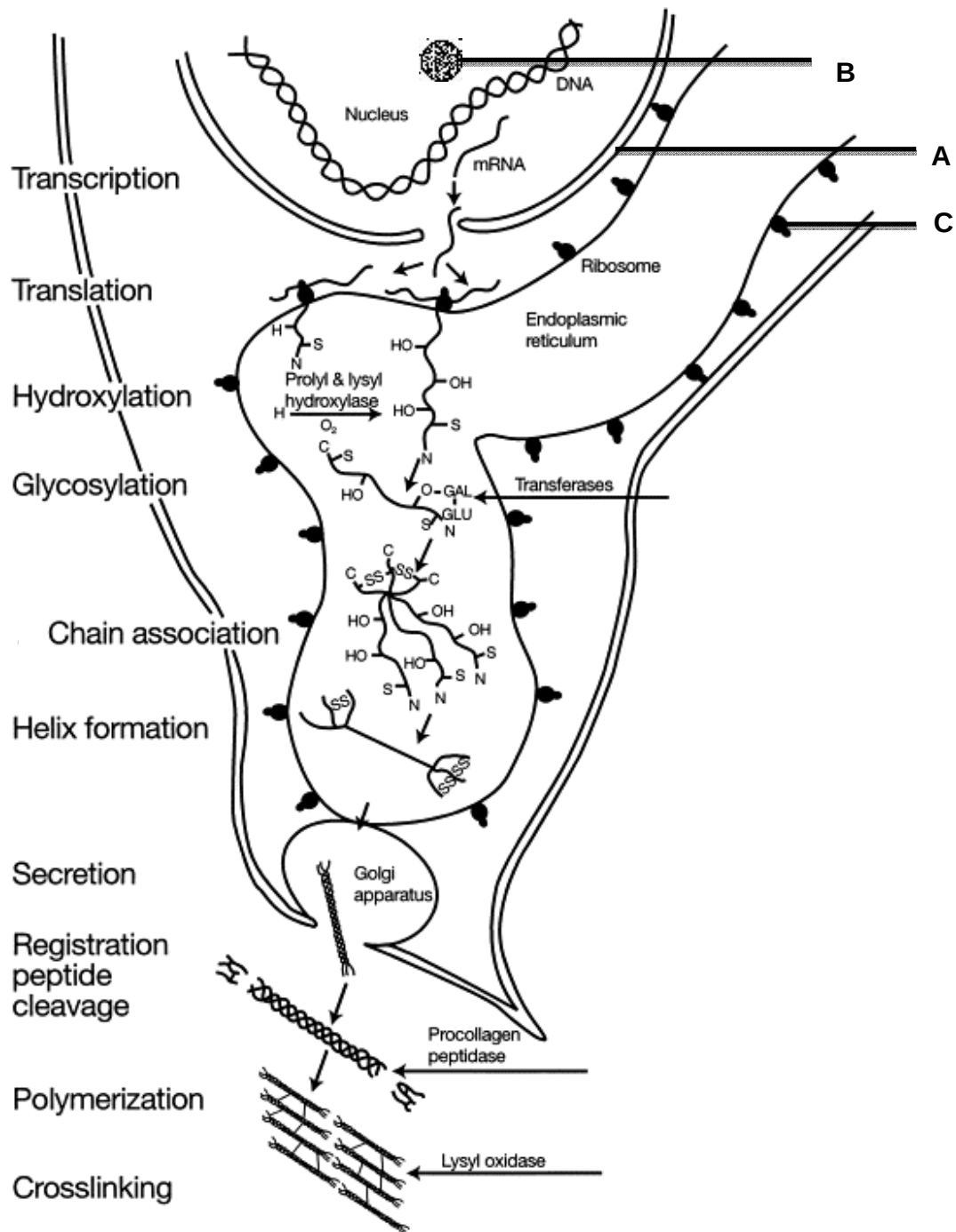


Fig. 1.1

(a)	(i)	State two reasons how A is important to the synthesis of collagen.	
			[2]
	(ii)	Explain how the functions of the two organelles, B and C, are linked.	[4]
	(b)	Suggest how modification such as hydroxylation in the rough endoplasmic reticulum results in collagen having a high tensile strength.	[2]
		• Result in the addition of <u>hydroxyl group/ polar groups</u> • that form <u>hydrogen bonds</u>; • Chain association to form <u>triple helix/ three polypeptides</u> strengthens the tropocollagen;	
	(c)	A molecule of messenger RNA (mRNA) was produced during the transcription of the collagen gene. Part of the template sequence of DNA was ATGC. Fig. 1.2 shows the part of the molecule of messenger RNA corresponding to that sequence of four bases.	

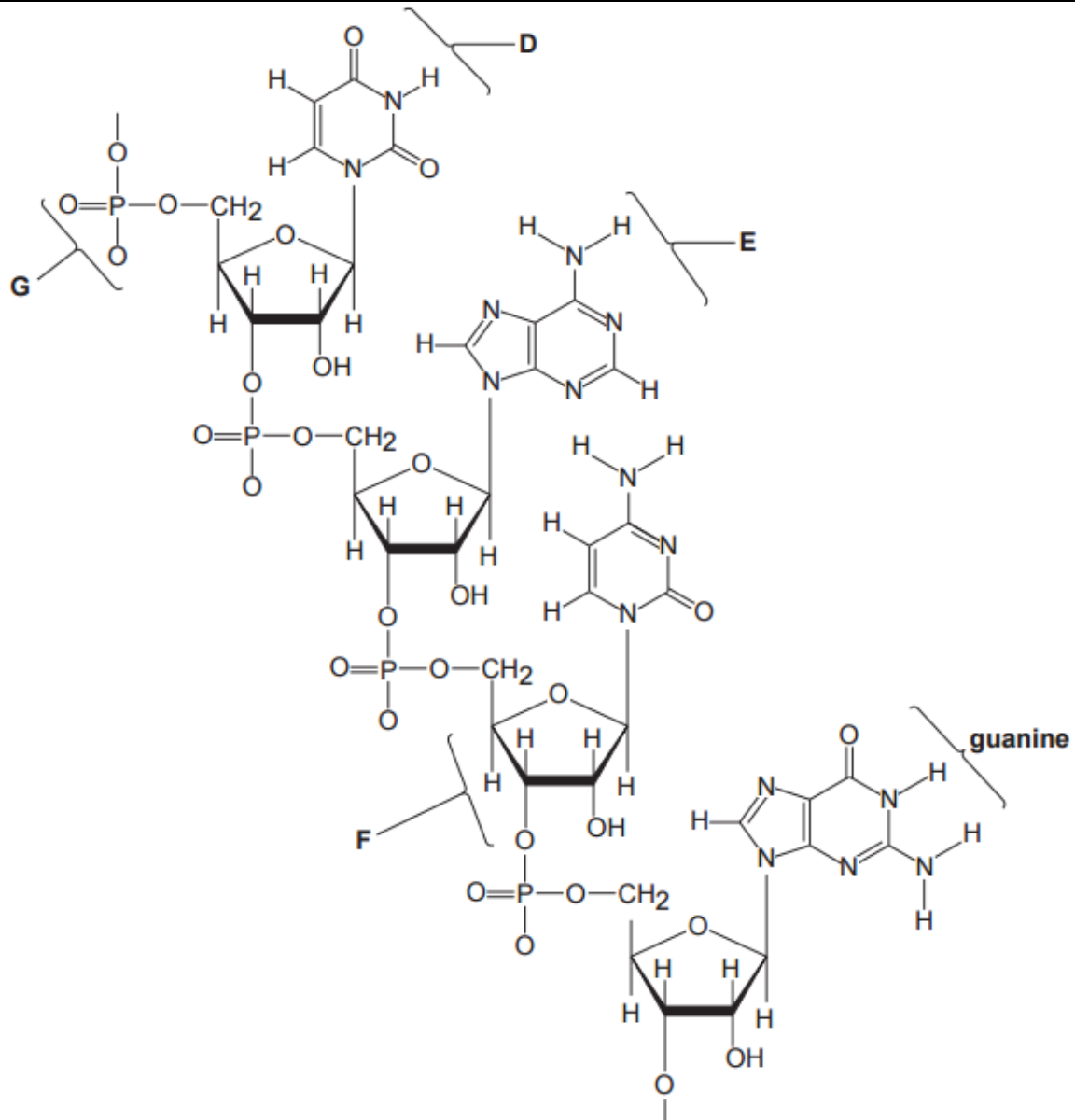


Fig. 1.2

Name the parts of the mRNA molecule shown in **Fig. 1.2** labelled **D**, **E**, **F** and **G**.

D

E

F

G

[2]

[Total: 10 marks]

2

Fig. 2.1 shows a virus found in a sample of water.

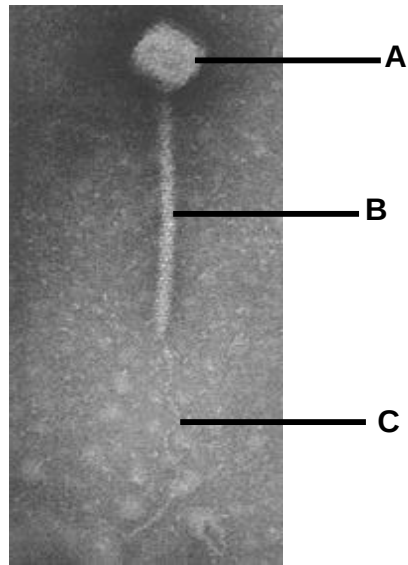


Fig. 2.1

(a) State the name of structures A – C.

A

B

C

[3]

(b) Lambda virus undergoes lysogenic pathway after infecting its host cell. Distinguish between the lytic and lysogenic pathway.

.....

.....

.....

[2]

			The following passage is an extract taken from an online article: “For the new study, the researchers developed a particle that works in a similar way - it targets and kills specific bacteria, without causing the cells to burst and release their toxins. They call the particles "phagemids" because they infect the target bacteria with plasmids - small DNA molecules that can copy themselves inside cells. Using synthetic biology, the team engineered the plasmids to express proteins and peptides - short-chain amino acids - that are toxic to the bacterial host cell. The toxins are designed to disrupt key cell processes such as replication, with the effect that the bacterial cell dies without bursting. The team systematically tested a variety of peptides and toxins and showed how, when some are combined in the phagemids, they kill the great majority of bacterial cells within a culture. The method they have developed is highly targeted.” - Medical News Today, 1st July 2015 http://www.medicalnewstoday.com/articles/296127.php	
	(c)	(i)	Suggest how these toxins mentioned in the passage may work.	
				[2]
		(ii)	Explain why “the method described is highly targeted”.	
				[1]

		(iii)	Give one advantage and one disadvantage of using phagemids to kill infectious bacteria.	
				[2]
			[Total: 10 marks]	

3 Iron ion concentration can be kept constant by transferrin membrane receptor proteins.

At high intracellular iron ion concentration, transferrin receptor proteins are inactive. At low intracellular iron ion concentration, transferrin receptor proteins will transfer and increase iron ions into the cytoplasm.

Production of transferrin receptors are regulated at the mRNA level by active IRE-binding proteins (IRPs) which bind to an iron response element (IRE) that forms a hairpin loop on the 3' untranslated region (3' UTR) of transferrin mRNA as shown in Fig. 3.1.

Transferrin receptor proteins

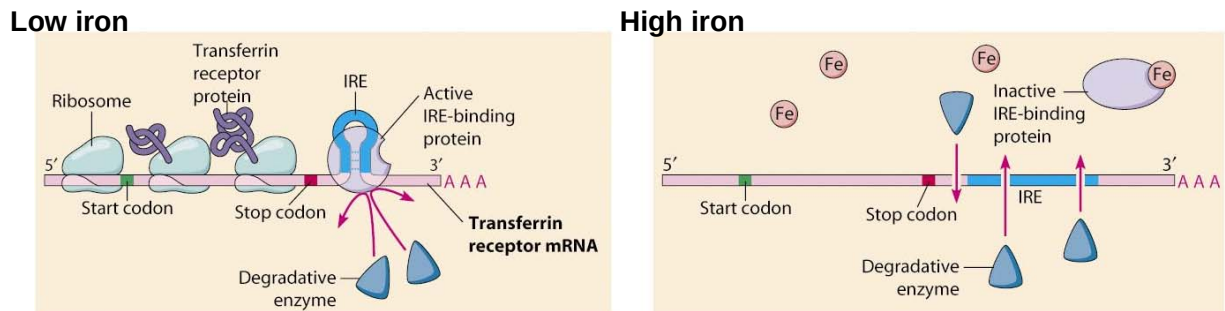


Fig. 3.1

(a) With reference to the structure of transferrin receptor mRNA in Fig. 3.1,

(i) explain how IRE can bind to the active IRP;

[2]

(ii) describe how expression of the transferrin receptor genes may be controlled at the **translational** level when iron ion concentration is **high**.

[3]

Cancer has always been associated with telomerase gene expression.

Human telomerase (hTERT) gene was expressed in a cancer cell, and the results are shown in Fig. 3.2.

	Tumor Volume (Mean, in mm ³)				
	10 days	13 days	17 days	20 days	24 days
Control	0	0	4.3	5.05	17.87
hTERT	38.07	115.7	146.29	228.98	319.56

	Tumor Growth (mm ³ /day)	Final Tumor Weight (Mean, in mg)
Control	1.19	Undetermined
hTERT	19.25	342

Fig. 3.2

(b) With reference to Fig. 3.2,

(i) describe the relationship between hTERT gene expression and cancer.

[2]

(ii) explain your results in **(b)(i)**.

[3]

[Total: 10 marks]

4	The <i>H-ras</i> oncogene codes for the <i>ras</i> protein, which is a type of G-protein. Fig. 4.1 shows part of the base sequence of a proto-oncogene and of the <i>H-ras</i> oncogene, which arises from the proto-oncogene by mutation. The site of this mutation is where GTPase enzyme of <i>ras</i> protein is located. Fig. 4.1 also shows part of the primary structure of their two encoded proteins. <table><tr><td colspan="14"></td><td colspan="2">proto-oncogene</td></tr><tr><td>ATG</td><td>ACG</td><td>GAA</td><td>TAT</td><td>AAG</td><td>CTG</td><td>GTG</td><td>GTG</td><td>GGC</td><td>GCC</td><td>GGC</td><td>GGT</td><td>GTG</td><td>GGC</td></tr><tr><td>met</td><td>thr</td><td>glu</td><td>tyr</td><td>lys</td><td>leu</td><td>val</td><td>val</td><td>gly</td><td>ala</td><td>gly</td><td>gly</td><td>val</td><td>gly</td></tr><tr><td colspan="14"></td><td colspan="2">oncogene</td></tr><tr><td>ATG</td><td>ACG</td><td>GAA</td><td>TAT</td><td>AAG</td><td>CTG</td><td>GTG</td><td>GTG</td><td>GGC</td><td>GCC</td><td>GTC</td><td>GGT</td><td>GTG</td><td>GGC</td></tr><tr><td>met</td><td>thr</td><td>glu</td><td>tyr</td><td>lys</td><td>leu</td><td>val</td><td>val</td><td>gly</td><td>ala</td><td>val</td><td>gly</td><td>val</td><td>gly</td></tr></table> Fig. 4.1															proto-oncogene		ATG	ACG	GAA	TAT	AAG	CTG	GTG	GTG	GGC	GCC	GGC	GGT	GTG	GGC	met	thr	glu	tyr	lys	leu	val	val	gly	ala	gly	gly	val	gly															oncogene		ATG	ACG	GAA	TAT	AAG	CTG	GTG	GTG	GGC	GCC	GTC	GGT	GTG	GGC	met	thr	glu	tyr	lys	leu	val	val	gly	ala	val	gly	val	gly	
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(a)	State the first three codons of mRNA of <i>ras</i> proto-oncogene. 	[1]																																																																																								
(b)	Explain how this mutation may change the structure and function of <i>ras</i> protein. 	[4]																																																																																								

Fig. 4.2 shows the cell cycle-stimulating pathway that involves *ras* protein.

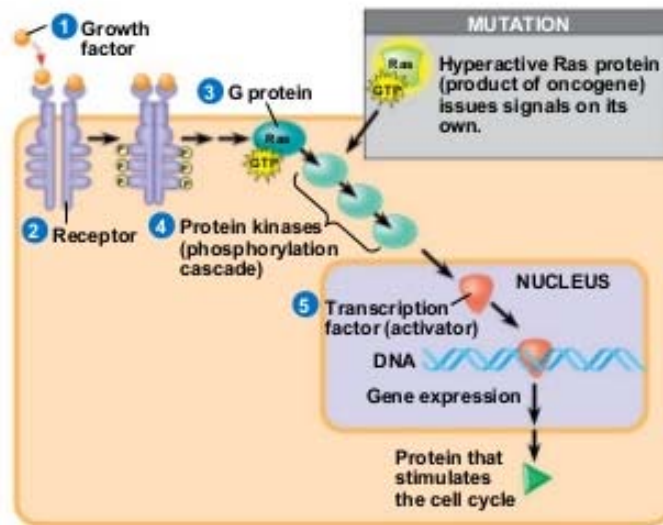


Fig. 4.2

(c) Using Fig. 4.2, describe how a mutated *ras* protein can stimulate cell cycle.

[3]

(d) It was found that a simple gene amplification of *ras* proto-oncogene is less likely to be oncogenic. Suggest why.

[2]

[Total: 10 marks]

5	(a)	Distinguish between the inheritance of multiple genes and multiple alleles.	
			[2]
		In cattle, coats may be solid white, solid black, or black-and-white-spotted. When true breeding solid whites are mated with true-breeding solid blacks, the F1 generation consists of all solid white individuals. Matings among the F1 generation have resulted in the following ratio: 23 solid white 6 black-and-white spotted 3 solid black	
	(b)	(i)	Is coat colour in cattle an example of multigenic inheritance or an example of multiple alleles at one gene locus?
			[1]
		(ii)	Explain the mode of inheritance observed.
			[2]

	(c)	Using a genetic diagram, predict the genotypes and phenotypes of a cross between a black-and-white spotted cow with a black parent, and a heterozygous solid white cow.	
			[3]
	(d)	Explain why it would be useful to carry out a chi-squared test on these results. No calculations are required to answer this question.	
			[2]
[Total: 10 marks]			

- 6** During the process of glycolysis, glucose is converted by a series of steps into two molecules of pyruvate. Fig. 6.1 outlines glycolysis.

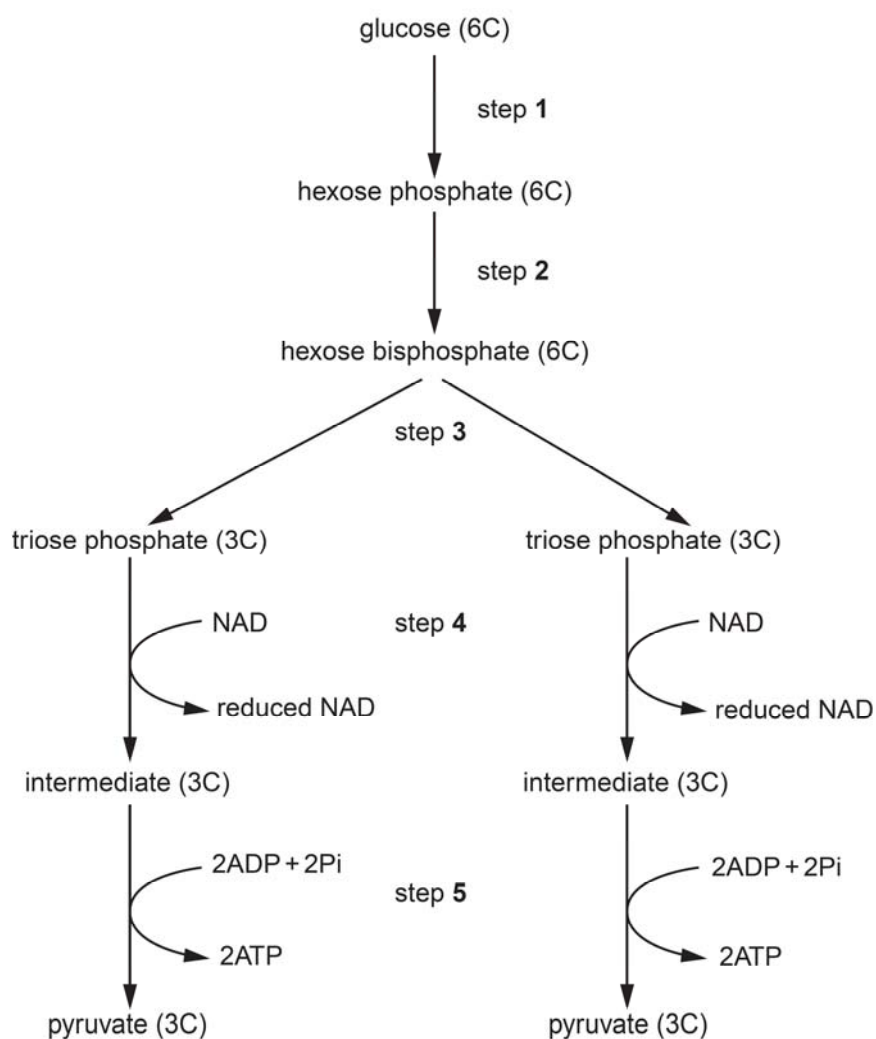


Fig. 6.1

- (a)** With reference to Fig. 6.1, state the reactions occurring at:

(i) steps 1 and 2

(i) step 3

(ii) step 4.

[2]

(b)	List three ways in which the synthesis of ATP in cytoplasm is different from the synthesis of ATP in mitochondria.	
		[3]
(c)	Explain how ATP is able to transfer energy in cells.	
		[3]

- (d) Fig. 6.2 outlines anaerobic respiration in yeast cells. Complete Fig. 6.2 by writing in the missing compounds.

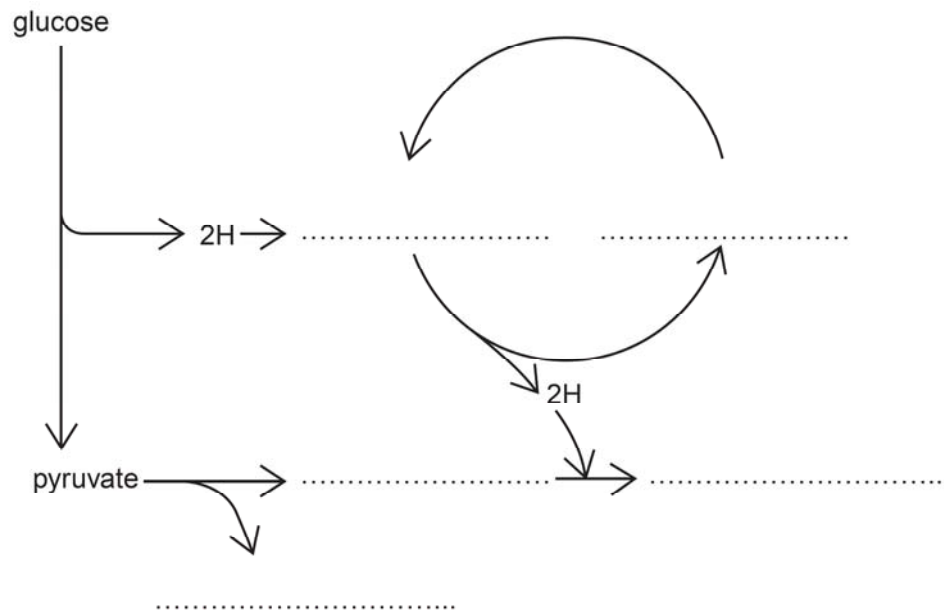


Fig. 6.2

[2]

[Total: 10 marks]

- 7 The evolution of haemoglobin molecules has been studied extensively by comparing the amino acid sequences in both myoglobin and haemoglobin. About 500 million years ago, a gene duplication event occurred and one copy became the present day myoglobin and the other evolved into an oxygen-transport protein that gave rise to the present day haemoglobin (Fig. 7.1).

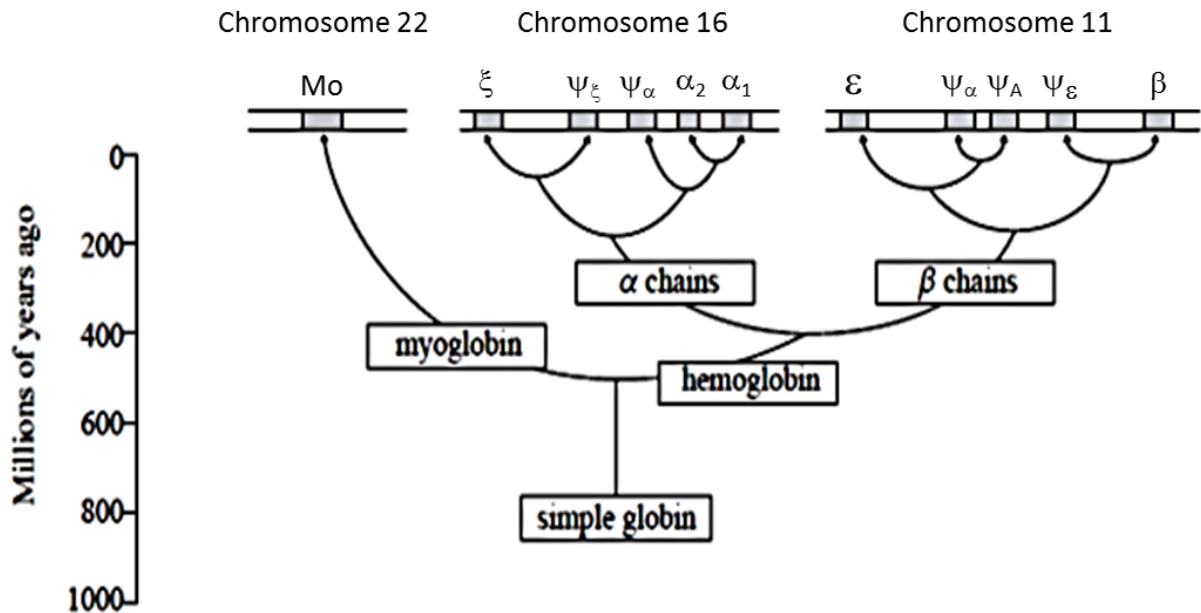


Fig. 7.1

- (a) State how many years ago the simple globin molecule diverged into myoglobin and haemoglobin.

.....

[1]

- (b) Suggest why changes observed in the sequence of amino acids may lead to an under-estimation of the actual number of mutations.

.....

[2]

	(c)	Explain how variations in the haemoglobin gene can be used as a molecular clock.	
			[3]
	(d)	Haemoglobin transports oxygen in red blood cells of all vertebrates such as fishes. The common name of <i>Gambusia affinis</i> is mosquitofish. Explain why the scientific name is better than the common names.	
			[2]
	(e)	Scientific names reflect the classification of organisms. Describe how classification differs from phylogeny.	
			[2]
			[Total: 10 marks]

8 Fig. 8.1 shows endings of neurones X and Y in the heart.

Stimulation of Y leads to an increase in the concentration of the secondary messenger cyclic AMP (cAMP), in the cytoplasm of the heart cell and the stimulation of X leads to a decrease.

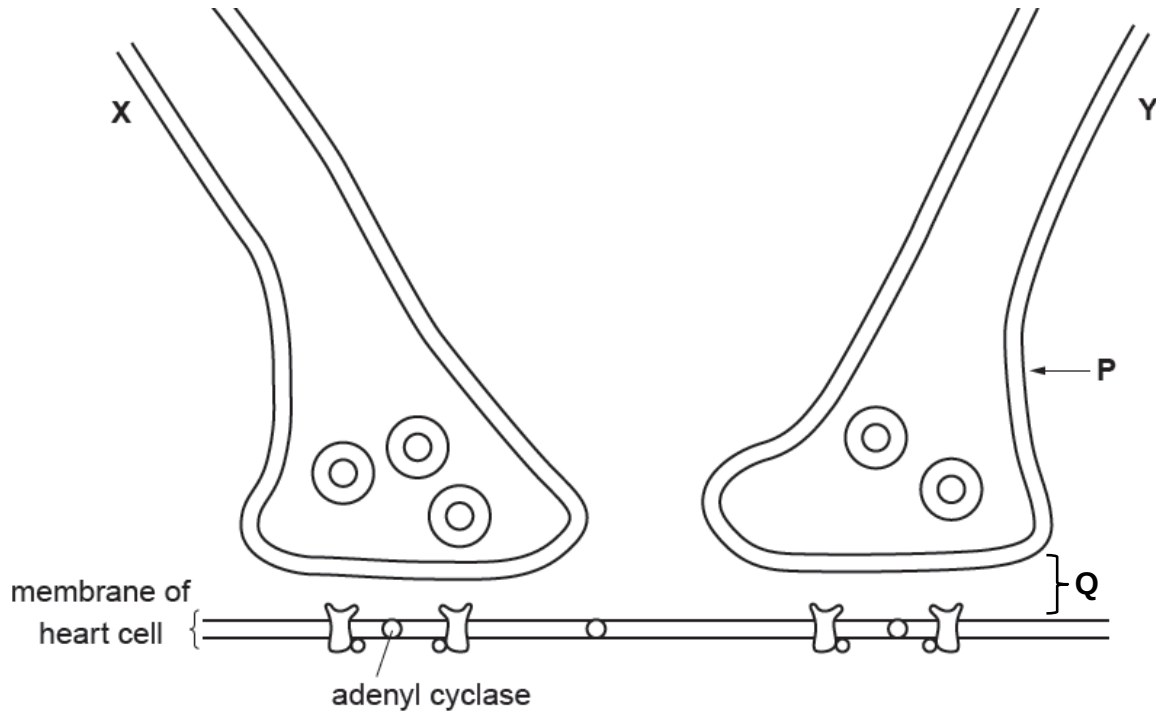


Fig. 8.1

(a) Describe the events that follow the arrival of an impulse at P on neurone Y that leads to an increase in cAMP.

[5]

	(b)	The intake of drugs such as cocaine alters nerve impulse transmission between neurones, causing an increase in heart rate. Suggest how this occurs.	[2]
	(c)	Distinguish between the impulse that arrives at P and the signal that crosses Q.	[3]
	[Total: 10 marks]		

Section B

Answer **EITHER 9 or 10.**

Write your answers on the separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section (a), (b) etc., as indicated in the question.

9	(a)	Using <i>lac</i> operon as an example, describe how prokaryotes such as <i>E.coli</i> is able to respond to changes in the environment.			[7]
	(b)	Describe the cellular responses that glucagon initiates in order to regulate blood glucose concentration.			[6]
	(c)	Compare triglyceride and starch as a storage molecule.			[7]
					[Total: 20]
Or					
10	(a)	Explain how fluidity of biological membranes can be maintained.			[7]
	(b)	Explain the importance of membrane to mitochondrion			[6]
	(c)	Discuss the factors that contribute to speciation.			[7]
					[Total:20]